

Proofs of Selected Examples from The Theory of
Partitions

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Chapter 1

Example 1

Proposition. The number of partitions of n in which each part appears two, three, or five times equals the number of partitions of n into parts congruent to 2, 3, 6, 9 or 10 modulo 12.

Lemma 1. A positive integer is congruent to 2, 3, 6, 9 or 10 modulo 12 iff it is an odd multiple of 2 or an odd multiple of 3.

Proof of Lemma 1. The odd residue classes modulo 12 are [1], [3], [5], [7], [9], and [11]. An odd multiple of 2 has the form $2m$, where m is odd. Thus the possible residue classes of $2m$ are

$$\begin{aligned} 2[1] &= 2[7] = [2], \\ 2[3] &= 2[9] = [6], \\ 2[5] &= 2[11] = [10]. \end{aligned}$$

An odd multiple of 3 has the form $3m$, where m is odd. Thus the possible residue classes of $3m$ are

$$\begin{aligned} 3[1] &= 3[5] = 3[9] = [3], \\ 3[3] &= 3[7] = 3[11] = [9]. \end{aligned}$$

Therefore, the odd multiples of 2 or 3 are exactly the integers congruent to 2, 3, 6, 9, or 10 modulo 12. ■

Proof of the Proposition. Let $p_\alpha(n)$ denote the number of partitions of n in which each part appears two, three, or five times. Let $p_\beta(n)$ denote the number of partitions of n into parts congruent to 2, 3, 6, 9, or 10 modulo 12. Thus

$$\sum_{n \geq 0} p_\alpha(n) q^n = \prod_{n \geq 1} (1 + q^{2n} + q^{3n} + q^{5n}),$$

and by Lemma 1

$$\sum_{n \geq 0} p_\beta(n) q^n = \prod_{n \geq 1} \frac{1}{(1 - q^{2(2n-1)})(1 - q^{3(2n-1)})}.$$

Finally

$$\begin{aligned} \sum_{n \geq 0} p_\beta(n) q^n &= \prod_{n \geq 1} \frac{1 - q^{4n}}{1 - q^{2n}} \cdot \frac{1 - q^{6n}}{1 - q^{3n}} \\ &= \prod_{n \geq 1} (1 + q^{2n}) (1 + q^{3n}) \\ &= \prod_{n \geq 1} (1 + q^{2n} + q^{3n} + q^{5n}) \\ &= \sum_{n \geq 0} p_\alpha(n) q^n. \end{aligned} \quad \blacksquare$$

Proof of the Proposition (q-Pochhammer). Let $p_\alpha(n)$ denote the number of partitions of n in which each part appears two, three, or five times. Let $p_\beta(n)$ denote the number of partitions of n into parts congruent to 2, 3, 6, 9, or 10 modulo 12. Thus

$$\sum_{n \geq 0} p_\alpha(n) q^n = (-q^2; q^2)_\infty (-q^3; q^3)_\infty,$$

and by Lemma 1

$$\sum_{n \geq 0} p_\beta(n) q^n = \frac{1}{(q^2; q^4)_\infty (q^3; q^6)_\infty}.$$

Finally

$$\begin{aligned} \sum_{n \geq 0} p_\beta(n) q^n &= \frac{(q^4; q^4)_\infty (q^6; q^6)_\infty}{(q^2; q^2)_\infty (q^3; q^3)_\infty} \\ &= (-q^2; q^2)_\infty (-q^3; q^3)_\infty \\ &= \sum_{n \geq 0} p_\alpha(n) q^n. \end{aligned}$$

■

Example 2

Proposition. The number of partitions of n in which only odd parts may be repeated, here denoted $p_o(n)$, equals the number of partitions of n in which no part is repeated more than three times, here denoted $p_3(n)$.

Proof.

$$\begin{aligned}
 \sum_{n \geq 0} p_o(n) q^n &= \prod_{n \geq 1} (1 + q^{2n}) \left(\frac{1}{1 - q^{2n-1}} \right) \\
 &= \prod_{n \geq 1} \left(\frac{1 - q^{4n}}{1 - q^{2n}} \right) \left(\frac{1}{1 - q^{2n-1}} \right) \\
 &= \frac{\prod_{n \geq 1} (1 - q^{4n})}{\prod_{n \geq 1} (1 - q^{2n})(1 - q^{2n-1})} \\
 &= \prod_{n \geq 1} \frac{1 - q^{4n}}{1 - q^n} \\
 &= \sum_{n \geq 0} p_3(n) q^n. \quad \blacksquare
 \end{aligned}$$

Proof (q-Pochhammer).

$$\begin{aligned}
 \sum_{n \geq 0} p_o(n) q^n &= (-q^2; q^2)_\infty (q; q^2)_\infty^{-1} \\
 &= \frac{(q^4; q^4)_\infty}{(q^2; q^2)_\infty (q; q^2)_\infty} \\
 &= \frac{(q^4; q^4)_\infty}{(q; q)_\infty} \\
 &= \sum_{n \geq 0} p_3(n) q^n. \quad \blacksquare
 \end{aligned}$$

Example 3

Proposition. The number of partitions of n in which only parts $\not\equiv 0 \pmod{2^m}$ may be repeated, here denoted $p_\alpha(n)$, equals the number of partitions of n in which no part appears more than $2^{m+1} - 1$ times, here denoted $p_\beta(n)$.

Proof.

$$\begin{aligned}
\sum_{n \geq 0} p_\alpha(n) q^n &= \prod_{n \geq 1} (1 + q^{2^m n}) \left(\frac{1 - q^{2^m n}}{1 - q^n} \right) \\
&= \prod_{n \geq 1} (1 + q^{2^m n}) (1 + q^n + \dots + q^{(2^m - 1)n}) \\
&= \prod_{n \geq 1} (1 + q^n + \dots + q^{(2^{m+1} - 1)n}) \\
&= \prod_{n \geq 1} \frac{1 - q^{2^{m+1} n}}{1 - q^n} \\
&= \sum_{n \geq 0} p_\beta(n) q^n. \quad \blacksquare
\end{aligned}$$

Proof (q-Pochhammer).

$$\begin{aligned}
\sum_{n \geq 0} p_\alpha(n) q^n &= (-q^{2^m}, q^{2^m}; q^{2^m})_\infty (q; q)_\infty^{-1} \\
&= \frac{(q^{2^{m+1}}; q^{2^{m+1}})_\infty}{(q; q)_\infty} \\
&= \sum_{n \geq 0} p_\beta(n) q^n. \quad \blacksquare
\end{aligned}$$

Example 4

Proposition. The number of partitions of n with unique smallest part and largest part at most twice the smallest part, here denoted $p_\alpha(n)$, equals the number of partitions of n in which the largest part is odd and the smallest part is larger than half the largest part, here denoted $p_\beta(n)$.

Scratchwork

The generating function for $p_\alpha(n)$ is

$$A(q) = \sum_{n \geq 1} \frac{q^n}{(q^{n+1}; q)_n},$$

and the generating function for $p_\beta(n)$ is

$$B(q) = \sum_{n \geq 1} \frac{q^{2n-1}}{(q^n; q)_n}.$$

Let the v^{th} term of $A(q)$ be denoted

$$a_v(q) = \frac{q^v}{(q^{v+1}; q)_v},$$

and let the v^{th} term of $B(q)$ be denoted

$$b_v(q) = \frac{q^{2v-1}}{(q^v; q)_v}.$$

Let $A_m(q)$ denote the finite series

$$A_m(q) = \sum_{n=1}^{2m-1} a_n(q),$$

and let $B_m(q)$ denote the finite series

$$B_m(q) = \sum_{n=1}^m b_n(q),$$

Conjecture 1. Then, for all $m \in \mathbb{Z}_+$,

$$A_m(q) \equiv B_m(q) \pmod{q^{2m}}. \quad (0.1)$$

Let

$$\rho_r(n, m) = \frac{1}{m} \sum_{j=0}^{m-1} e^{2\pi i j(n-r)/m} = \begin{cases} 1 & \text{if } n \equiv r \pmod{m}, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$[q^N]a_v(q) = \sum_{x_1+x_2+\dots+x_v=N-v} \prod_{i=1}^v \rho_0(x_i, v+i),$$

and

$$[q^N]A_m(q) = \sum_{v=1}^{2m-1} [q^N]a_v(q),$$

and

$$[q^N]b_v(q) = \sum_{x_1+x_2+\dots+x_v=N-(2v-1)} \prod_{i=1}^v \rho_0(x_i, v+i-1),$$

and

$$[q^N]B_m(q) = \sum_{v=1}^m [q^N]b_v(q).$$

Conjecture 2. Then, for all $0 \leq N < 2m$

$$[q^N]A_m(q) = [q^N]B_m(q). \quad (0.2)$$

Conjecture 1 and Conjecture 2 are equivalent.